

MODULI OF CURVES AND DEFORMATION THEORY

SERKAN SALIK

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1. INTRODUCTION

A moduli space $\mathcal{M}$ is a space whose points correspond to a certain class of geometric objects. Simple examples include $\mathbb{P}^n$, which represents the moduli space of lines in $\mathbb{C}^{n+1}$, or $G(k, n)$, which represents the moduli space of k -planes in $\mathbb{C}^n$. Some more sophisticated examples are the moduli space of genus g Riemann surfaces, or the moduli space of line bundles on a fixed Riemann surface X . We would also like $\mathcal{M}$ to have a topology, with nearby points in $\mathcal{M}$ corresponding to “nearby objects”.

The moduli space of a class of objects are interesting to study when those objects can vary continuously in families. For example, consider the family of elliptic curves E_λ , where E_λ is the zero locus of the polynomial $y^2 - x(x - 1)(x - \lambda)$. The polynomials are varying holomorphically with respect to some parameter $\lambda \in \mathbb{C}$, and correspondingly the spaces E_λ are getting holomorphically deformed. A non-example of this phenomenon occurs when we (instead of elliptic curves, or genus g curves in the holomorphic or algebraic category) consider closed surfaces in the smooth category. Here, the isomorphism class of a surface is uniquely determined by an integer g , its genus. Thus we can't have a continuous family of closed surfaces without all of them being isomorphic, which is obviously not interesting.

Now, suppose we want to construct a moduli space $\mathcal{M}_g$ of Riemann surfaces of genus g . What should it mean for nearby points on $\mathcal{M}_g$ to correspond to nearby Riemann surfaces of genus g ? A natural definition of this is that there should exist a *universal fibration* $\mathcal{C}_g \rightarrow \mathcal{M}_g$, i.e. a complex

manifold¹ $\mathcal{C}_g$, a surjective holomorphic map $\pi: \mathcal{C}_g \rightarrow \mathcal{M}_g$, such that the fiber above a point $X \in \mathcal{M}_g$ is (isomorphic to) the curve X . Moreover, suppose we have a family $F \rightarrow B$, meaning complex manifolds F, B , a surjective holomorphic map $\pi: F \rightarrow B$, such that each fiber is a curve of genus g . Then, there should exist maps $\varphi: F \rightarrow \mathcal{C}_g, B \rightarrow \mathcal{M}_g$, with φ a fiberwise isomorphism such that the following diagram commutes.

$$\begin{array}{ccc} F & \xrightarrow{\varphi} & \mathcal{C}_g \\ \downarrow & & \downarrow \\ B & \longrightarrow & \mathcal{M}_g \end{array}$$

Unpacking this definition, we see that we need the existence of a map $B \rightarrow \mathcal{M}_g$ such that the fiber above $b \in B$ is isomorphic to the fiber above the image of b in $\mathcal{M}_g$ (in addition to the map on total spaces being a fiberwise isomorphism). This motivates the following definition.

Definition 1.1. A *fine moduli space of curves of genus g* , denoted by $\mathcal{M}_g$, is a complex manifold together with a universal fibration $\mathcal{C}_g \rightarrow \mathcal{M}_g$ (also a complex manifold) satisfying the above properties.

Let's do an example, or at least an almost-example.

Example 1.2. Let $g = 1$. Each genus 1 curve is isomorphic to $\mathbb{C}/L$ for some lattice L , and by rescaling we could assume $L = \langle 1, \tau \rangle$ for some τ in the upper half plane $\mathbb{H}$. Moreover, if $\tau' \in \mathbb{H}$ differed from τ by an element of $SL_2(\mathbb{Z})$, ie. $\tau' = \frac{a\tau+b}{c\tau+d}$ for some $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$, then $\mathbb{C}/\langle 1, \tau \rangle \cong \mathbb{C}/\langle 1, \tau' \rangle$. Thus, a natural candidate for $\mathcal{M}_1$ is $\mathbb{H}/SL_2(\mathbb{Z})$. Now, we have to construct a universal family over this space, but we must first see if this space is actually a manifold. Unfortunately, it's not! It is an orbifold with two singularities, meaning it locally looks like $\mathbb{R}^n$ modulo a finite group action.

In fact, for $g \geq 1$, a fine moduli space for genus g curves never exists! This is caused by the presence of nontrivial automorphisms. This isn't just a heuristic, and can be made into an actual proof.

Proposition 1.3. *Fine moduli spaces $\mathcal{M}_g$ don't exist for $g \geq 1$.*

Proof. Suppose they did. It's clear that for each g , there are curves of that genus with non-trivial automorphisms. For example, take the hyperelliptic curve given by $y^2 = p(x)$ with $\deg p = 2g + 1$, which has a non-trivial involution $(x, y) \mapsto (x, -y)$. So, let X be a genus g curve with a non-trivial automorphism φ , and take a cylinder $C = S^1 \times [0, 1]$. Consider $C \times X / \sim$, where $(0, x) \sim (1, \varphi(x))$. This can be given a complex structure by expressing it as a quotient of $\mathbb{C} \times X / \mathbb{Z}^2$, where (m, n) acts on (z, p) via $(z + m + ni, \varphi^m(p))$. This gives us a (topologically) nontrivial family over a torus $\mathbb{T}$,

¹these will be orbifolds, but for the sake of exposition let's ask for manifolds

with each fiber isomorphic to X . Thus, by the properties of a fine moduli space, there exists maps f, g with the following diagram commuting.

$$\begin{array}{ccc} F & \xrightarrow{g} & \mathcal{C}_g \\ \downarrow & & \downarrow \\ \mathbb{T} & \xrightarrow{f} & \mathcal{M}_g \end{array}$$

However, since all the fibers of F are isomorphic, the map f must be constant, and thus the family F must be a product $\mathbb{T} \times X$. But F was topologically non-trivial, thus we have a contradiction. ■

In fact, the situation for tori is even worse, since tori have continuous families of automorphisms given by translations. This is dealt with by introducing the moduli space of curves with *marked points*, which keeps track of the data of a select number of distinguished points on a curve. Unfortunately, we won't explore this further in this paper.

1.1. Riemann's Computation. In this section, we'll pretend that fine moduli spaces $\mathcal{M}_g$ exist and are complex manifolds, and we'll do some semi-rigorous computations. In particular, we'll compute the dimension of $\mathcal{M}_g$, which was first computed by Riemann to be $3g - 3$ for $g \geq 2$. Of course, he didn't have the machinery we have today to say this is the dimension of a complex manifold, so he said that Riemann surfaces (or rather, "Abelian functions") depend on $3g - 3$ parameters. Here is his computation.

He first considered the *Hurwitz space* $\text{Hur}_{d,g}$, which is the space of pairs (X, f) , where X is a genus g curve and f is a degree d branched cover of $\mathbb{P}^1$. Here we are being a bit imprecise about what we are quantifying X over, so just suppose the moduli space has been constructed and that everything we want works as expected. This admits a natural fibering over $\mathcal{M}_g$, given by taking a pair (X, f) to X . The fiber is then the space of degree d holomorphic maps $X \rightarrow \mathbb{P}^1$, call it $W(X, d)$. So, we have a fibration

$$\begin{array}{ccc} W(X, d) & \longrightarrow & \text{Hur}_g(X, d) \\ & & \downarrow \\ & & \mathcal{M}_g \end{array}$$

If these spaces were complex manifolds, we would have $\dim \mathcal{M}_g = \dim \text{Hur}_g(X, d) - \dim W(X, d)$. Let's compute the dimensions of these latter two spaces. The dimension of $W(X, d)$ is easy: the space of degree d maps with a given zero locus $D = p_1 + \dots + p_d$ is just $h^0(D)$, and since we can vary each of the p_i , we should have $\dim W(X, d) = d + h^0(D)$. For large d , $h^1(D) = 0$, so $h^0(D) = d - g + 1$. Thus $\dim W(X, d) = 2d - g + 1$.

Now, to compute the dimension of the Hurwitz space. First, let us compute how many critical points a typical map f in this space has. By Riemann-Hurwitz, we have $\chi(X) = d\chi(\mathbb{P}^1) - b$, so $b = 2d + 2g - 2$. One can then show that for a dense subset of points B in $\text{Sym}_b(\mathbb{P}^1)$, there is a finite

number of branched covers with this branching locus, and so $\dim \text{Hur}_{d,g} = 2d + 2g - 2$. Thus, we see $\dim \mathcal{M}_g = 2d + 2g - 2 - (2d - g + 1) = 3g - 3$.

2. CONSTRUCTION OF MODULI SPACES

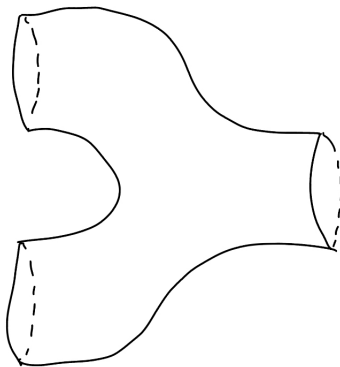
We'll now outline different approaches to constructing moduli spaces. We'll also mention different ways to compute their dimension and tangent spaces, and how to understand deformation of complex structures.

2.1. Almost Complex Structures Modulo Gauge Equivalence. It is a theorem that every almost complex structure on a Riemann surface integrates to a unique genuine complex structure on X . Two almost complex structures J, J' determine isomorphic Riemann surfaces if there is a diffeomorphism ϕ of X such that $J' = d\phi \circ J \circ d\phi^{-1}$. Thus the moduli space of Riemann surfaces should just be $\mathcal{J}/\text{Diff}^+(M)$, where $\mathcal{J}$ is the space of positively oriented almost complex structures and $\text{Diff}(X)$ is the space of orientation preserving diffeomorphisms of X (we're assuming here that X is an oriented smooth surface).

2.2. Teichmuller Theory and Hyperbolic Geometry. Here, we use the fact that a conformal class, a hyperbolic metric, and an almost complex structure all uniquely determine each other. Thus, the moduli space of curves should be isomorphic to the moduli space of hyperbolic structures, which can be seen, as above, to be $\mathcal{H}/\text{Diff}^+(X)$, where by $\mathcal{H}$ I mean the space of hyperbolic metrics with the C^∞ topology. Note that this space is isomorphic to the space obtained by first quotienting by the group of diffeomorphisms isotopic to the identity, $\text{Diff}_0^+(X)$, and then quotienting by the *mapping class group*, which is the group of diffeomorphisms of a surface modulo isotopy. It can be shown that the *Teichmuller space* $\mathcal{H}/\text{Diff}_0^+(X)$ is naturally homeomorphic to $\mathbb{R}^{6g-6}$. So if this were a complex manifold, the dimension would be $3g - 3$, which would agree with Riemann's computation!

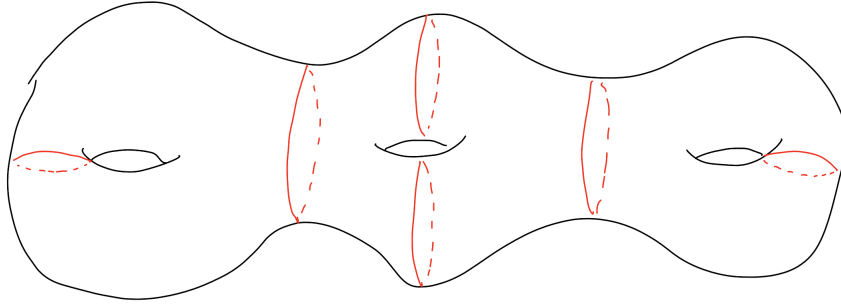
The way to see directly that the moduli space is of real dimension $6g - 6$ by taking a short detour into hyperbolic geometry.

A *pair of pants* is something that looks as follows.



It is a theorem from hyperbolic geometry that for each triple a, b, c of nonnegative integers, there is a unique hyperbolic metric on the pair of pants such that the boundaries are geodesics and have length a, b, c . If one of these is zero, this means the boundary forms a sort of cusp. Now, if we have a pair of pants with boundary lengths a, b, c , and a pair of pants with boundary lengths a, b', c' , we can glue the boundaries of length a to obtain a new hyperbolic metric. This is because in coordinates, a hyperbolic metric looks like the standard hyperbolic plane, and a geodesic in the hyperbolic plane can be made to just be a straight vertical line. Then, the problem is reduced to gluing two regions in the hyperbolic plane with boundary a straight vertical line, and this can obviously be done to give a new hyperbolic surface.

It is a fact that each genus g surface splits up into $3g - 3$ pairs of pants.



Note that the pants at the ends of the surface are sort of degenerate, in that two of their boundary components are glued together. Each choice of lengths for the red cuts we made gives a different hyperbolic surface, modulo the mapping class group. Naively, this would lead us to conclude that the isometry class of the surface depends on $3g - 3$ parameters. However, we can also pick the angle at which we glue each pair of red cuts, and this gives another $3g - 3$ parameters to give a net total of $6g - 6$ parameters.

2.3. Geometric Invariant Theory. This approach is more algebraic. Let X be a subvariety of $\mathbb{P}^N$, and let $I \subset \mathbb{C}[x_0, \dots, x_N]$ be the ideal of vanishing polynomials on X . Let $I^{(n)}$ denote the set of homogeneous polynomials of degree n in I , and similarly let $\mathbb{C}^{(n)}[x_0, \dots, x_N]$ be the degree n homogeneous polynomials in $N + 1$ variables. Then, for large n , a theorem of Hilbert says that $p(n) := \dim \mathbb{C}^{(n)}[x_0, \dots, x_N] / I^{(n)}$ is a polynomial in n , called the *Hilbert polynomial* of X .

We will construct the moduli of curves as a *Hilbert scheme*, which means a moduli space of subvarieties of $\mathbb{P}^N$ with a fixed Hilbert polynomial. Indeed, let $Q_N(n)$ be the dimension of the space of homogeneous polynomials of degree n in $N + 1$ variables. We can see from the stars-and-bars method in combinatorics that $Q_N(n) = \binom{N+n}{N}$. Now, there is the key fact that there exists for each Hilbert polynomial $p(n)$ an integer n_0 such that for any subvariety X with Hilbert polynomial $p(n)$ and for any integer $n > n_0$, we have $\dim \mathbb{C}^{(n)}[x_0, \dots, x_N] / I^{(n)}[x_0, \dots, x_N] = p(n)$. Thus fix a Hilbert

polynomial $p(n)$, an integer $n > n_0$. Then $I_X^{(n)}$ determines a plane of dimension $Q_N(n) - p(n)$ in the space $\mathbb{C}^{(n)}[x_0, \dots, x_n]$ of dimension $Q_N(n)$. Thus, to each subvariety $X \subset \mathbb{P}^N$ we get an element of the Grassmannian $G(Q_N(n), Q_N(n) - p(n))$. The subset of planes obtained in such a way is precisely the *Hilbert scheme* $\text{Hilb}_N^{p(n)}$.

So far we've only constructed some sort of moduli space, which is not necessarily the moduli space of curves of fixed genus g . It turns out that this problem can be solved by using so-called *pluricanonical embeddings*. Indeed, each base-point free line bundle induces a map to some projective space, so consider the maps induced by the third tensor powers of the canonical bundle. One computes easily from Riemann-Roch (for $g \geq 2$, of course) that $h^0(\Omega^{\otimes 3}) = 5g - 5$, and so $\Omega^{\otimes 3}$ induces a map to $\mathbb{P}^{5g-6}$. It is a fact that this is an embedding, and the image is a subvariety with Hilbert polynomial $p_g(n) = 3(2g - 2)n - g + 1$. It turns out all curves in $\mathbb{P}^{5g-6}$ with this Hilbert polynomial are 3-canonical curves of genus g . Moreover, any two isomorphic curves will differ by a projective transformation when considered as subvarieties of $\mathbb{P}^{5g-6}$ under this pluricanonical embedding. This is because if they were isomorphic, this would induce an isomorphism between their canonical bundles, and therefore also between the space of global sections of the third tensor power of their canonical bundles. Thus the only ambiguity in identifying them is the choice of identification of the space $\Gamma(\Omega^{\otimes 3})^*$ with $\mathbb{C}^N$, and this is taken care of by modding out by the projective linear group.

Thus, our moduli space will be given by taking the Hilbert scheme $\text{Hilb}_{5g-6}^{p_g(n)}$, considering the open and dense subspace of smooth varieties (because there might be some singular ones) and quotienting out by $PGL(5g-6)$. The only problem is that we don't know what kind of singularities this quotient will have, and for all we know it might not even be Hausdorff. Ensuring this is the task of geometric invariant theory.

Let us see why the quotient by a large group action could result in a non-Hausdorff space. Suppose a Lie group G acts on some vector space V , and let $M \subset V$ be a G -invariant subset. The quotient will look very bad if two orbits Gx and Gy have closures that intersect, as this would imply M/G is not Hausdorff. Two orbits have closures that intersect if and only if both of them contain the origin. This motivates the following definition.

Definition 2.1. A point $x \in M$ is called *stable*² if $\overline{Gx}$ does not contain the origin.

At first sight, it doesn't seem easy to tell when this would be the case. So let us do a simple example, when $G = \mathbb{C}^\times$ and $V = \mathbb{C}^2$. In this case, any such action can be decomposed into two 1-dimensional actions of the form $\lambda \cdot v = \lambda^w v$, where w is called the *weight* of the action. Thus, the action will take the form $\lambda \cdot v = (\lambda^{w_1} v, \lambda^{w_2} v)$. If the w_i have the same sign, then as $\lambda \rightarrow 0$ or $\lambda \rightarrow \infty$, $\lambda \cdot v$ will tend to zero and thus 0 will be contained in the closure of the orbit of v . But

²I'm not sure if this is the standard definition, since it disagrees with Wikipedia, where this is called semi-stable. But this is the definition used in the book [KLP18].

if they have opposite signs, the subset of points of $\mathbb{C}^2$ that aren't on either coordinate axis stay a bounded distance from 0 and hence are stable. Thus, we see that this action contains an open and dense *stability domain*, i.e. an open and dense subset of $\mathbb{C}^2$ is stable under this action. Even though this may seem very special, there is a generalization to arbitrary actions.

Theorem 2.2 (Hilbert-Mumford Numerical Criterion). *A point $x \in V$ is stable with respect to some action $G \curvearrowright V$ if for every 1-parameter subgroup of G , the expansion of x in terms of the positive and negative eigendirections of γ contains nonzero coordinates with positive and negative weights.*

Finally, the construction of the moduli space of curves of genus $g \geq 2$ proceeds as follows.

- (1) Construct the Hilbert scheme $\text{Hilb}_{5g-6}^{p_g(n)}$. This is embedded into the Grassmannian $G(Q_N(n), Q_N(n) - p_g(n))$.
- (2) Embed the Grassmannian via the Plücker embedding into some large projective space $\mathbb{P}^K$. Consider the preimage of the Grassmannian under the projection $\mathbb{C}^{K+1} - 0 \rightarrow \mathbb{P}^K$. The action of the projective linear group on this Grassmannian can then be extended to an action on $\mathbb{P}^K$, and thus to a linear action on $\mathbb{C}^{K+1}$.
- (3) Use the Hilbert-Mumford criterion to verify that the points in the Hilbert scheme corresponding to smooth 3-canonical curves are stable with respect to the action of the above group. Therefore, the quotient is a smooth orbifold.
- (4) The above steps can all be accompanied with the construction of a universal curve.

3. COMPUTING THE TANGENT SPACE OF $\mathcal{M}_g$ AND DEFORMATION THEORY

Intuitively, a tangent vector at a point $\Sigma \in \mathcal{M}_g$ should consist of an infinitesimal holomorphic deformation of Σ . A *deformation* of Σ is a holomorphic fibration $E \rightarrow \Delta$, where Δ is the unit disk and the fiber above 0 is Σ . Intuitively, we need to take the “principal linear part” or “first order part” of this deformation to get an infinitesimal deformation. We'll illustrate a few ways of doing this in the following subsections.

3.1. Variation of Transition Maps. This is the most down to earth approach. Take a cover $\{U_\alpha\}$ of a surface X by coordinate charts, and consider the transition maps $g_{\alpha\beta}$, which completely determine the isomorphism class of X . A *deformation* is then a 1-parameter family of transition maps $\{g_{\alpha\beta}(t)\}$, which satisfy the cocycle condition at each time t . The infinitesimal information encoded by this genuine deformation is $g'_{\alpha\beta}(t) \in T'X$, i.e. a 1-cocycle valued in the holomorphic tangent bundle of X . Now, the cocycles that are coboundaries are those that come from trivial deformations. This is because they come from a Čech 0-cochain, and this would correspond to an infinitesimal variation of coordinates on each of the U_α . Assuming we can realize each 1-cocycle as the cocycle associated to a deformation, or just taking on faith that these should correspond

to “infinitesimal deformations”, we see that the space of infinitesimal deformations of X should naturally be identified with $H^1(TX)$.

This gives us another way of computing $\dim \mathcal{M}_g$. Indeed, since $TX = \Omega^{-1}$, we know $\deg(TX) = 2 - 2g$. Then, by Riemann-Roch, $h^0(TX) - h^1(TX) = 2 - 2g - g + 1 = 3 - 3g$. However, $h^0(TX) = 0$ since TX has negative degree, and multiplying by -1 on both sides we get $h^1(TX) = 3g - 3$.

3.2. Abstract, Sheaf Theoretic Approach. Before we go on, let us note that the appearance of cohomology in describing infinitesimal deformations, as [Don11] says, is a very general phenomenon. The simplest example is given by associative real algebras. In this context, the object whose deformations we’re considering is simply a bilinear, associative map $M: \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n$. The set of such M is acted on in the obvious way by $GL(n, \mathbb{R})$. Thus, if we let $\mathcal{A}$ be the set of bilinear associative maps $\mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n$, the moduli space of associative real algebras of dimension n would be $\mathcal{A}/GL(n, \mathbb{R})$.

Now, let’s see how to compute the space of infinitesimal deformations of an element $M_0 \in \mathcal{A}$. Write $M_0(x, y) = xy$, and consider an infinitesimal deformation $M_0 + \varepsilon$, where ε is some bilinear map $\mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n$. Then the equation requiring $M_0 + \varepsilon$ to be associative is of the form $\varepsilon(x, y)z + \varepsilon(xy, z) - \varepsilon(x, yz) - x\varepsilon(y, z) + Q(\varepsilon)$, where $Q(\varepsilon)$ is quadratic in ε . So taking the linearized equation, the space of deformations is given by the set of solutions to the equation $\varepsilon(x, y)z + \varepsilon(xy, z) - \varepsilon(x, yz) - x\varepsilon(y, z) = 0$. But there are some trivial deformations, and those are given by actions of $1 + \delta \in GL(n, \mathbb{R})$, i.e. infinitesimal linear automorphisms. We see that the action of $1 + \delta$ takes M_0 to M_1 , with $M_1(x, y) = xy + \delta(x)y + x\delta(y) - \delta(xy) + O(\delta^2)$. Thus, linearizing again, we expect that the space of non-trivial infinitesimal deformations is given by the space of solutions to the equation $\varepsilon(x, y)z + \varepsilon(xy, z) - \varepsilon(x, yz) - x\varepsilon(y, z) = 0$ modulo the space of solutions of the form $\varepsilon(x, y) = \delta(x)y + x\delta(y) - \delta(xy)$, as these correspond to trivial deformations given by the action of an element of $GL(n, \mathbb{R})$. Aha! This is beginning to sound more and more like cohomology. Reformulating, if we let $\mathcal{A}_p$ denote the space of $(p + 1)$ -multilinear maps $\mathbb{R}^n \times \cdots \times \mathbb{R}^n \rightarrow \mathbb{R}^n$, define $d_0: \mathcal{A}_0 \rightarrow \mathcal{A}_1$ by $d_0(\delta)(x, y) = \delta(x)y + x\delta(y) - \delta(xy)$ and $d_1: \mathcal{A}_1 \rightarrow \mathcal{A}_2$ by $d_1(\varepsilon)(x, y, z) = \varepsilon(x, y)z + \varepsilon(xy, z) - \varepsilon(x, yz) - x\varepsilon(y, z)$. Our previous discussion shows that the space of nontrivial infinitesimal deformations of M_0 is given by $\frac{\ker d_1}{\text{im } d_0}$.

Returning to our original discussion, we now describe a more abstract approach to deformation of Riemann surfaces. Here, we think of a complex structure on a surface Σ as being given by a sheaf of rings $\mathcal{O}_\Sigma$. We define an infinitesimal deformation of Σ to consist of the following:

- A sheaf of rings S on Σ
- A global section $t \in \Gamma(\Sigma, S)$ with $t^2 = 0$,
- An isomorphism of sheaves of rings $\alpha: S/(tS) \rightarrow \mathcal{O}_\Sigma$

We say two infinitesimal deformations S and S' are equivalent if there is an isomorphism of sheaves that preserves the above structure, i.e. takes t to t' and is compatible with the isomorphisms α and

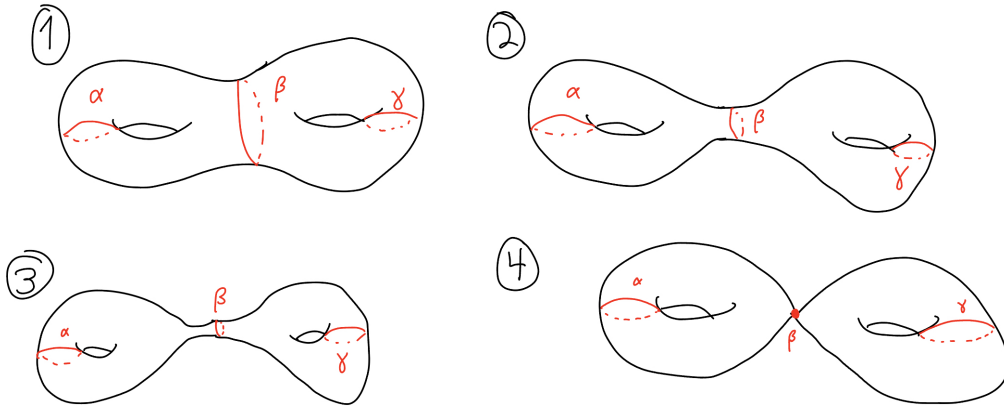
α' . The trivial deformation is given by $S = \mathcal{O}_\Sigma + \mathcal{O}_\Sigma t$, with the obvious ring structure. We'll make it part of the definition that locally, every deformation is equivalent to this one.

Why should this abstract definition correspond to our intuitive understanding of what a deformation is? [Don11] motivates this as follows. Suppose we had a genuine deformation $\pi: \mathcal{X} \rightarrow \Delta$, where Δ is the unit disc, $\mathcal{X}$ is a complex surface, and the fiber above zero is the Riemann surface Σ we are considering. We then obtain a sheaf of rings $\tilde{S}$ on Σ , obtained by restriction of holomorphic functions on $\mathcal{X}$. This contains the element t which is the pullback of the coordinate function on Δ . Now, around each point in Σ , we have a holomorphic trivialization of this fibration, say on an open set U . Then each point in $\mathcal{O}(U)$ is represented as $\sum_i f_i t^i$, but this depends on the choice of coordinates. However, the first order part of this expansion, which can be thought of as the image of $\sum_i f_i t^i$ under the quotient $\tilde{S} \rightarrow \tilde{S}/t^2 \tilde{S}$, is canonical. Intuitively, this represents the infinitesimal deformation of the sheaf of holomorphic functions on U , and amounts to considering, as Donaldson says, functions on $\mathcal{X}$ up to first order in the transverse coordinate t . We then have the following sheaf-theoretic fact, which relates back to the approach through deformation of transition maps of coordinate charts. We won't give a proof of it, however.

Proposition 3.1. *The equivalence classes of infinitesimal deformations of Σ are in natural $1-1$ correspondence with classes in $H^1(T\Sigma)$.*

4. COMPACTIFICATIONS OF $\mathcal{M}_g$

The moduli space $\mathcal{M}_g$ we've constructed above is not compact. One way to see this, using the hyperbolic geometry picture of $\mathcal{M}_g$, is to visualize a sequence of genus g curves that have one of the cuts in the pairs of pants shrinking.



Here, we shrink the curve labeled β , until it shrinks to a Riemann surface with a *node*, i.e. a point that locally looks like $wz = 0$ in $\mathbb{C}^2$. Clearly no subsequence of these converges, because all of them would be converging to this nodal curve, which isn't in the moduli space to begin with.

A more algebraic way to see the process of degeneration shown above with hyperbolic geometry is to consider the family of curves E_λ defined by $y^2 = x^3 + x^2 + \lambda$ (this is genus 1 now, but the same problem is happening). This is non-singular for λ small and not equal to zero, but when $\lambda = 0$ this is a nodal cubic.

The reason this is a problem is because we often want to do intersection theory on the moduli space of curves (or the moduli space of curves in a Kähler manifold), and intersection theory only works well when we have compactness. Thus, we would like to find a compactification $\overline{\mathcal{M}}_g$ of $\mathcal{M}_g$, and preferably, we would like the new points we add to represent something geometrically meaningful. The above picture gives us a clue: the points we add to compactify should be some sort of degenerate curves, and hopefully they will have reasonable singularities. These will be *stable curves*.

Definition 4.1. A *stable curve* is a nodal curve with finite automorphism group.

Example 4.2. We give some examples of stable curves.

- Take two elliptic curves E_1 and E_2 with some basepoints p, q and glue them along these points. The automorphism group of this new curve is generated by the set of automorphisms of E_1 preserving p , of E_2 preserving q , and possibly of the isomorphism $E_1 \rightarrow E_2$ that take p to q . The number of these maps is clearly finite.
- Take an elliptic curve E , two points $p, q \in E$, and identify them. The group of automorphisms of this new curve is the set of automorphisms preserving the pair of points p and q , which is again finite.

Then, the so-called *Deligne-Mumford compactification* of $\mathcal{M}_g$, denoted by $\overline{\mathcal{M}}_g$, consists of biholomorphic equivalence classes of stable curves. To finish, we'll state some results about this space.

Theorem 4.3 (Deligne-Mumford). *For any $g \geq 2$, there exists a family $\overline{\mathcal{C}}_g \rightarrow \overline{\mathcal{M}}_g$ of stable curves of genus g such that for any other family of stable curves of genus g , the obvious diagram commutes. The space $\mathcal{M}_g$ is defined uniquely up to biholomorphism and is irreducible.*

Here, by a family of stable curves we mean a map of complex orbifolds $\mathcal{X} \rightarrow B$ with each fiber being a stable curve modulo its automorphism group.

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